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Psychometric properties of a Romanian translation of the compensatory eating and behaviors in response to alcohol consumption scale (CEBRACS)

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Abstract
The 21-item Compensatory Eating and Behaviors in Response to Alcohol Consumption Scale (CEBRACS) is the most widely used instrument for the assessment of food and alcohol disturbance. In the present study, we examined the psychometric properties of a novel Romanian translation of the CEBRACS in two studies. In Study 1 with 1,022 Romanian university students, exploratory factor analysis and parallel analysis indicated that our data reduced to three latent factors reflecting Alcohol Effects, Bulimia, and Diet and Exercise motives. In Study 2 with 614 Romanian university students, we found that a 3-factor model based on exploratory structural equation modelling (ESEM) had better fit to the data than a model based on confirmatory factor analysis. In both studies, we found evidence of complete measurement invariance across gender identity and a lack of differential item functioning based on participant age and body mass index. The results of Study 2 also supported the convergent and concurrent validity of the Romanian CEBRACS, insofar as the three factors were significantly associated with symptoms of disordered eating, problematic alcohol use, body image, and life satisfaction. These results suggest that a 3-factor, ESEM-based model of the Romanian CEBRACS has adequate psychometric properties.

Keywords Food and alcohol disturbance · Drunkorexia · Alcohol · Romania · Test adaptation · Psychometrics

Introduction

Food and alcohol disturbance (FAD) – colloquially termed "drunkorexia" (Chambers, 2008) – refers to a set of behaviours involving a functional relationship between disordered eating and alcohol use (Berry et al., 2024; Choquette et al., 2018b). Although definitions of FAD vary widely (Shepherd et al., 2023), it is broadly thought to involve the use of disordered eating-related behaviours (e.g., restricting, purging, or excessive exercise) to enhance desired outcomes (i.e., enhancing intoxication levels) or to mitigate undesirable outcomes (i.e., to compensate for calories consumed from drinking alcohol). In Westernised countries, such as the United States, France (Choquette et al., 2018a), and the United Kingdom (Bradbury et al., 2025), FAD is highly prevalent, with over half of young adults engaged in at least one FAD behaviour. Such findings are of growing concern because of the detrimental impact of alcohol use and disordered eating behaviours (Looby et al., 2025), as well as

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the overlap of risk for substance abuse and eating disorders (Hunt & Forbush, 2016).

Although various quantitative instruments have been developed to measure FAD (for a review, see Berry et al., 2024), the most commonly used is the 21-item Compensatory Eating and Behaviors Related to Alcohol Consumption Scale (Rahal et al., 2012). The CEBRACS measures the frequency with which an individual engages in a variety of behaviours to compensate for alcohol-related calories or to enhance the effects of alcohol across three time-points: before, during, and after alcohol use. Based on the results of a principal components analysis with data from college students from the United States, Rahal et al. (2012) proposed a 4-factor model consisting of one alcohol-related motivation factor (Alcohol Effects) and three caloric-reduction factors (Bulimia, Dietary Restraint and Exercise, and Restriction). Despite this, most studies using the CEBRACS have utilised total scores (Berry et al., 2024), which likely obfuscates what is being measured by this instrument.

Despite its widespread usage, there has been some criticism of the instrument's factorial validity. For example, using confirmatory factor analysis (CFA), several studies with college students in the United States failed to find adequate fit of the parent 4-factor model (Choquette et al., 2020; Meinerding et al., 2024). Based on the results of a subsequent exploratory factor analysis (EFA), Choquette et al. (2020) proposed an alternative scoring structure based on time-points: a 2-factor model for items capturing FAD prior to alcohol consumption, a 3-factor model for items related to FAD during alcohol consumption, and a 2-factor model for items related to post-consumption. Conversely, based on an EFA, Meinerding et al. (2024) extracted a 19-item, 3-factor model (Alcohol Effects, Diet/Exercise, and Bulimia), but a subsequent CFA found that this 3-factor model had poor fit to the data.

Meinerding et al. (2024) also examined CFA-based fit of alternative models, including a unidimensional model and a 2-factor model (Alcohol Enhancement Effects and Caloric Reduction), but these consistently had poor fit to the data. Based on these results, Meinerding et al. (2024) have called for use of the CEBRACS to be discontinued, whereas other authors have recommended revisions to this instrument to improve its psychometric properties (Berry et al., 2024). A further complication to these issues is the possibility that the CEBRACS is measuring constructs that are more common in English-speaking, Westernised countries. For example, controlled and supervised access to alcohol in family settings in some non-English-speaking European countries has been associated with more responsible drinking and lower risk of alcohol abuse in later life (Chentsova et al., 2024). To date, however, only two studies have examined the factor structure of the CEBRACS in non-English-speaking samples.

The two studies used EFAs to examine the factor structure of the CEBRACS in Italian adolescents (Pinna et al., 2015) and French young adults (Ritz et al., 2023), respectively. Both studies were able to extract a primary Alcohol Effects factor, identical to that extracted in the parent study. Beyond this, however, the extracted factor structures diverged. A further four factors were extracted in the Italian sample (Laxative Use, Dietary Restraint and Exercise, Diuretic Use, and Restraint and Vomiting) (Pinna et al., 2015). Conversely, in the French sample, Ritz et al. (2023) extracted an additional three factors (Dietary Restraint and Exercise, Purging, and Vomiting and Extreme Fasting), which showed item-level variations from the parent factor structure. Across both studies, factor scores were also found to be moderately associated with problematic alcohol use and weakly associated with symptoms of disordered eating.

The present studies

In view of these equivocal findings, there is a need to further examine the factorial validity and psychometric properties of the CEBRACS in diverse populations. To this end, the present study examined the psychometric properties of a novel Romanian translation of the CEBRACS. Romania is a useful site in which to consider the factor structure of the CEBRACS because, following a period of transition from a centralised to more market-oriented economy, Romania has experienced a rapid transformation in its food and alcohol consumption practices (Vintilă et al., 2020). Among Romanian young adults in particular, per capita alcohol consumption now surpasses that of most other European countries (Roșioară et al., 2024) and is associated with a range of negative outcomes (e.g., lower levels of physical activity and greater consumption of fast food; Năsui et al., 2021a, b). To date, however, we are not aware of any study that has assessed FAD in the Romanian context, which may reflect the lack of validated tools to do so.

To assess the factorial validity of the Romanian CEBRACS, we followed recent psychometric recommendations in adopting an EFA-to-CFA/exploratory structural equation modeling (ESEM) approach (Swami et al., 2023a)[1]. More specifically, in Study 1, we examined the factor structure of the CEBRACS using EFA as a first-pass analytic step, which allowed us to consider the most suitable model of the CEBRACS for our data without imposing any modeling constraints (i.e., a data-driven approach; Swami & Barron, 2019). In Study 2, we then cross-validated the optimal model by comparing the fit of CFA- and ESEM-derived models. CFA only allows items to load on to their respective hypothesised latent factor and forces loadings to be zero (i.e., it assumes "pure" latent factors; Morin, 2023). However, this is an unrealistic assumption, especially as previous EFAs have shown that cross-loadings on the CEBRACS are non-trivial (Choquette et al., 2020; Meinerding et al., 2024).

ESEM, on the other hand, allows for a relaxation of the zero cross-loadings requirement of CFA and thus provides a more realistic account of item behaviour (Morin, 2023; Swami et al., 2023a, b). While we allowed for an exploratory approach to hypothesising in Study 1, we preliminarily expected that an ESEM model would offer better fit to the data than a CFA model in Study 2. In addition, in both studies, we also assessed gender invariance of the CEBRACS (i.e., the extent to which the latent CEBRACS factors are equivalent across women and men) and the extent to which the CEBRACS items may function differently based on respondent age and body mass index (BMI). As a preliminary hypothesis, we expected that the optimal CEBRACS model would achieve full measurement invariance across gender in both studies and would show no differential item functioning (DIF).

In addition to an assessment of factorial validity, we also examined the construct validity of the Romanian CEBRACS based on the availability of additional instruments that have been validated for use in Romanian populations. Specifically, in Study 2, we assessed the convergent and concurrent validity of the CEBRACS. In terms of the former, we examined associations between CEBRACS factors and symptoms of disordered eating and problematic alcohol use, with the expectation of weak-to-moderate, positive associations. To assess concurrent validity, we additionally examined associations with body appreciation, self-esteem, and life satisfaction. In terms of concurrent validity, we expected that CEBRACS factors would be show weak-to-moderate negative correlations with these constructs.

Study 1

Participants

The initial sample for Study 1 consisted of 1,174 undergraduate students from several universities in Romania. However, after excluding participants who failed to correctly answer attention check items ($n=152$), we retained a final sample of 1,022 respondents, of whom 733 identified as women and 289 identified as men. Participants ranged in age from 18 to 45 years ($M=21.84$, $SD=2.09$) and in BMI from 15.24 to 45.17 kg/m^2 ($M=23.02$, $SD=4.10$).

Materials

Food and alcohol disturbance Participants completed a novel Romanian translation of the 21-item Compensatory Eating and Behaviors Related to Alcohol Consumption Scale (Rahal et al., 2012). This instrument is self-administered and is designed to measure FAD prior to, during, and after alcohol consumption over the past three months (Rahal et al., 2012). To translate the CEBRACS into Romanian, we followed a combined approach integrating back-translation and expert committee methods (see Swami & Barron, 2019). Initially, two bilingual individuals proficient in both Romanian and English translated the CEBRACS, its instructions, and response options from English into Romanian, and then back-translated it into English. An expert committee, composed of six authors of this study and an independent expert, then reviewed the translations and resolved any minor discrepancies to ensure accuracy and clarity. All items were rated on a 5-point scale, ranging from 1 (*never*; Romanian: *niciodată*) to 5 (*almost all the time*; Romanian: *aproape tot timpul*). The items of the CEBRACS in Romanian and English are reported in Table S1 in the Supplementary Materials.

Demographics Participants were asked to self-report their gender identity, age, height, and weight. Participant BMI was calculated as kg/m^2.

Procedures

The research protocol was developed following the 2013 Declaration of Helsinki and was approved by the ethics panel at the first author's institution (approval no. 2901/19.11.2024). All data were collected using Google Forms between January and February 2025. We employed a snowball recruitment technique, initially distributing the survey to various undergraduate student groups from Psychology programmes that provided extra course credits for participation. These participants were then asked to invite other undergraduate students from other faculties in Romania. In order to obtain a culturally homogeneous sample, we limited participation to students who were fluent in Romanian and who held Romanian citizenship and residency. Participants were informed that all data would remain anonymous and confidential, and that they could withdraw from the study at any point. We embedded two attention-check items (e.g., "What is the result of 2+2?") into the questionnaire and provided responses using a Likert-type scale consistent with whatever scale they were incorporated into. We excluded participants after failing one item. To support the collection of high-quality data, we restricted survey submissions to one per IP address (to avoid duplicate entries). The survey did not include any mandatory items, but participants were prompted to attend to missing imputations before leaving a survey page. All participants received written debriefing information upon completion of the survey package.

Statistical analyses

Mplus 8.11's was used to perform EFAs (Muthén & Muthén, 2024). The estimation used was the robust weighted least squares estimator with mean and variance adjusted statistics (WLSMV). No missing data were observed for CEBRACS responses. As per Hair and colleagues (2009), data factorability was assessed using the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy (which should ideally be ≥0.80) and Bartlett's test of sphericity (which should be significant). Firstly, ten EFAs correlated latent factors were estimated (Models 1–1 to 1–10) using Mplus's ESEM capabilities. We retained ten as the upper limit based on the fact that the maximum number of factors found with the CEBRACS was seven. Based on previous recommendations (e.g., Marsh et al., 2009), an oblique geomin rotation and an epsilon value of 0.5 were used to estimate the EFAs. Parallel analysis was used to determine the optimum number of factors to retain in this model. This test was conducted using the *Paran* (50 randomly generated data sets) package v.1.5.4 (Dino, 2025) in *R* v.4.5.2 (R Development Core Team, 2014). In addition to parallel analysis, fit indices were examined for each of the models, as recommended by Swami et al. (2021). Values ≥ 0.90 or > 0.95 for the comparative fit index (CFI) and the Tucker-Lewis index (TLI) indicated acceptable and good fit to the data, respectively (Hu & Bentler, 1999). Additionally, values ≤ 0.08 or < 0.06 for the root mean square error of approximation (RMSEA) indicated acceptable and good fit to the data, respectively (Hu & Bentler, 1999). The composite reliability of latent factors from the CEBRACS was estimated using McDonald's (1970) ω.

Secondly, the best EFA solution was retained to examine its measurement invariance as a function of gender identity. The following sequence was tested (Morin et al., 2011): (i) configural invariance; (ii) loadings invariance; (iii) thresholds strong; (iv) uniquenesses invariance; (v) invariance of the latent variances/covariances; and (vi) invariance of latent mean factors. Each model was contrasted to the previous one based on changes (Δ) in fit indices (i.e., CFI, TLI, and RMSEA). As recommended in the literature (e.g., Chen, 2007; Cheung & Rensvold, 2002), a model was considered as non-invariant when ΔCFI/ΔTLI decreases was higher than 0.01 and/or ΔRMSEA increases was higher than 0.015.

Thirdly, a multiple indicators multiple causes model (MIMIC; Morin, 2023; Swami et al., 2023a) was used to determine the presence of DIF and latent mean differences as a function of participants' age and BMI (i.e., standardized prior to the analyses to facilitate interpretation). As recommended (Morin, 2023; Swami et al., 2023a), these tests were performed using the following models: (a) null effects (all paths from age and BMI to the latent factors and items' thresholds were constrained to 0); (b) saturated (all paths from age and BMI to the latent factors were constrained to 0 and those to items' thresholds were free); and (c) factors only (paths from age and BMI to the latent factors were all free, while those to items' thresholds were constrained to 0). We considered that there were associations between predictors and item responses when improvement of fit indices (increases of ΔCFI/ΔTLI higher or equal to 0.01, and decreases of ΔRMSEA higher or equal to 0.015) were observed between the null effects models and both the saturated and Factors only models (Morin, 2023; Swami et al., 2023a). Finally, there is DIF when an improvement of fit indices is observed between the saturated model relative to the factors-only model.

Results and discussion

Exploratory factor analysis Both Bartlett's test of sphericity, $\chi^2(210) = 15{,}152.64$, $p < .001$, and the Kaiser-Meyer-Olkin measure of sampling adequacy (0.91), indicated that the CEBRACS items had adequate common variance for factor analysis with the data from Study 1. Table 1 presents the fit indices of the ten EFA models that were tested (Models 1–1 to 1–10). Results showed that fit indices for the unidimensional model of the CEBRACS (Model 1–1) was unsatisfactory (RMSEA > 0.09). On the other hand, models with three to nine factors (Models 1–2 and 1–9) all had acceptable-to-good fit to the data (CFI and TLI > 0.95; RMSEA ≤ 0.07). The EFA solutions (see Table 1) revealed a substantial improvement of fit indices (CFI and/or TLI >. 01) between the 2- and 3-factor models (Models 1–2 and 1–3) and a less substantial improvement in fit indices between the 3- and 4-factor models (Models 1–3 and 1–4). Further, a plateau in fit indices was reached for the 5-factor solution (Models 1–5). Therefore, based on fit indices, the most optimal representation of the data appears to be either the 3- or 4-factor models.

Examination of EFA models indicated that the 3-factor solution consisted of a primary factor mirroring that of the Alcohol Effects factor from Rahal et al. (2012; 7 items). The second factor grouped together the six items of the Bulimia subscale from Rahal et al. (2012) and a third factor included Diet and Exercise items, similar to that reported by Meinerding et al. (2024; 8 items). In contrast, the 4-factor solution comprised the first two dimensions (i.e.,

Table 1 Goodness-of-Fit statistics for the CEBRACS

Models	Study	N°	Description	$W\chi^2$	*df*	CFI	TLI	RMSEA	RMSEA 90% CI		CM	$\Delta W\chi^2$	*df*	*p*	ΔCFI	ΔTLI	ΔRMSEA
									LB	UB							
EFA	1	1–1	1 factor	1924.273	189	0.947	0.941	0.095	0.091	0.099	-	-	-	-	-	-	-
		1–2	2 factors	953.893	169	0.976	0.970	0.067	0.073	0.072	-	-	-	-	-	-	-
		1–3	3 factors	592.774	150	0.986	0.981	0.054	0.049	0.058	-	-	-	-	-	-	-
		1–4	4 factors	344.807	132	0.993	0.990	0.040	0.035	0.045	-	-	-	-	-	-	-
		1–5	5 factors	199.992	115	0.997	0.995	0.027	0.021	0.033	-	-	-	-	-	-	-
		1–6	6 factors	125.923	99	0.999	0.998	0.016	0.005	0.024	-	-	-	-	-	-	-
		1–7	7 factors	107.112*	84	0.999	0.998	0.016	0.003	0.025	-	-	-	-	-	-	-
		1–8	8 factors	78.344	70	1.000	0.999	0.011	0.000	0.022	-	-	-	-	-	-	-
		1–9	9 factors	53.696	57	1.000	1.000	0.000	0.000	0.017	-	-	-	-	-	-	-
		1–10	10 factors	NA	NA	NA	NA	NA	NA	NA	-	-	-	-	-	-	-
ESEM: MI across gender	1	2 − 1	Men	246.216*	150	0.991	0.987	0.047	0.036	0.057	-	-	-	-	-	-	-
		2–2	Women	345.204*	150	0.993	0.990	0.042	0.036	0.048	-	-	-	-	-	-	-
		2–3	Configural invariance	748.973*	300	0.988	0.983	0.054	0.049	0.059	-	-	-	-	-	-	-
		2–4	Weak invariance	647.195*	354	0.992	0.991	0.040	0.035	0.045	2–3	43.461	54	0.846	−0.004	+0.008	−0.014
		2–5	Strong invariance	757.531*	406	0.991	0.990	0.041	0.037	0.036	2–4	152.806	52	<0.001	−0.001	−0.001	+0.001
		2–6	Strict invariance	935.038*	427	0.987	0.987	0.048	0.044	0.052	2–5	178.677	21	<0.001	−0.004	−0.003	+0.007
		2–7	Variances-covariances invariance	775.064*	433	0.991	0.991	0.039	0.035	0.044	2–6	14.067	6	0.029	+0.004	+0.004	−0.009
		2–8	Latent means invariance	888.821*	436	0.988	0.989	0.045	0.041	0.049	2–7	36.150	3	<0.001	−0.003	−0.002	+0.006
DIF: Age & BMI	1	3 − 1	MIMIC Null effects	536.404*	192	0.990	0.986	0.042	0.038	0.046	-	-	-	-	-	-	-
		3 − 2	MIMIC Saturated	626.927*	150	0.986	0.976	0.056	0.051	0.060	3 − 1	106.384	42	<0.001	−0.004	−0.010	+0.014
		3–3	MIMIC Factors only	600.396*	186	0.987	0.983	0.047	0.043	0.051	3 − 1	33.297	6	<0.001	−0.003	−0.003	+0.005
CFA	2	4 − 1	3 factors	797.355*	186	0.983	0.981	0.073	0.068	0.078	-	-	-	-	-	-	-
ESEM		4 − 2	3 factors	460.978*	150	0.991	0.988	0.058	0.052	0.064	-	-	-	-	-	-	-
ESEM: MI across gender	2	5 − 1	Men	297.351*	150	0.991	0.988	0.064	0.053	0.075	-	-	-	-	-	-	-
		5 − 2	Women	212.132*	150	0.997	0.996	0.033	0.022	0.043	-	-	-	-	-	-	-
		5 − 3	Configural invariance	501.362*	300	0.995	0.993	0.047	0.039	0.054	-	-	-	-	-	-	-
		5 − 4	Weak invariance	526.015*	354	0.995	0.994	0.040	0.032	0.047	5 − 3	79.985	54	0.012	0.000	+0.001	−0.007
		5–5	Strong invariance	598.812*	407	0.995	0.995	0.039	0.032	0.046	5 − 4	96.118	53	<0.001	0.000	+0.001	−0.001
		5–6	Strict invariance	592.656*	428	0.996	0.996	0.035	0.028	0.042	5–5	23.339	21	0.326	+0.001	+0.001	−0.004
		5–7	Variances-covariances invariance	555.699*	434	0.997	0.997	0.030	0.022	0.037	5–6	13.106	6	0.041	+0.001	+0.001	−0.005
		5–8	Latent means invariance	611.709*	437	0.995	0.995	0.036	0.029	0.043	5–7	17.642	3	<0.001	−0.002	−0.002	+0.006
DIF: Age & BMI	2	6 − 1	MIMIC Null effects	466.117*	192	0.992	0.990	0.048	0.043	0.054	-	-	-	-	-	-	-
		6 − 2	MIMIC Saturated	448.320*	150	0.992	0.986	0.057	0.051	0.063	6 − 1	122.181	42	<0.001	0.000	−0.004	+0.009
		6 − 3	MIMIC Factors only	474.096*	186	0.992	0.989	0.050	0.045	0.056	6 − 1	36.131	6	<0.001	0.000	−0.001	+0.002

CEBRACS = Compensatory Eating and Behaviors in Response to Alcohol Consumption Scale; EFA = exploratory factor analysis; $W\chi^2$ = robust weighed least square (WLSMV) chi-square; *df* = degrees of freedom; CFI = comparative fit index; TLI = Tucker-Lewis index; RMSEA = root mean square error of approximation; 90% CI = 90% confidence interval of the RMSEA; LB = lower bound; UB = upper bound; CM = comparison model; Δ = change from the previous model; $\Delta W\chi^2$ = WLSMV chi square difference test (calculated with the Mplus DIFFTEST function); NA = not available; MI = measurement invariance; DIF = differential item functioning; MBI = body-mass index; MIMIC = multiple indicators multiple causes. CFA = confirmatory factor analyses; ESEM = exploratory structural equation modelling. * $p \leq .01$

Alcohol Effects and Bulimia), plus two additional factors splitting items from the third dimension (the two with 4 items). Additionally, this latter factor solution comprised a higher number of items with high cross-loadings. Finally, parallel analysis indicated that the the 3-factor model was the optimal solution (see Figure S1 in the Supplementary Materials). Therefore, based on the combination of fit indices, analyses of the factor structure, and results from the parallel analysis test, the 3-factor model was retained for subsequent analyses.

The standardized parameters and composite reliability coefficients for the 3-factor solution are reported in Table 2. All three factors had moderate-to-high loadings (Alcohol Effects: $\lambda=0.787$ to 0.876; $M_\lambda = 0.826$; Bulimia: $\lambda=0.435$ to 0.937; $M_\lambda = 0.791$; Diet and Exercise: $\lambda=0.495$ to 0.833; $M_\lambda = 0.695$), with small cross-loadings ($M_{|\lambda|} = 0.133$), with the exception of Item #4. Indeed, Item #4 tended to have a high level of association with the Alcohol Effects subscale. Finally, the results also revealed that the latent correlations between the three factors were significant, positive, and modest (all ≤ 0.460, $M_r = 0.407$) and that the composite reliability coefficients of the three factors were all satisfactory (Alcohol Effects: $\omega = 0.973$; Bulimia: $\omega = 0.948$; Diet & Exercise: $\omega = 0.939$; $M_\omega = 0.953$).

Table 2 Standardised parameters estimates from the 3-Factor exploratory factor analytic representation of the CEBRACS in the sample from study 1

Items	Alcohol effects (λ)	Bulimia (λ)	Diet & exercise (λ)	δ
CEBRACS1	**0.787**	0.201	<u>*0.055*</u>	0.184
CEBRACS2	−0.089	0.230	**0.722**	0.345
CEBRACS3	**0.876**	0.180	<u>*−0.017*</u>	0.112
CEBRACS4	0.401	0.197	**0.495**	0.237
CEBRACS5	<u>*0.111*</u>	**0.845**	<u>*0.053*</u>	0.164
CEBRACS6	**0.838**	0.192	0.062	0.094
CEBRACS7	**0.846**	<u>*0.023*</u>	0.164	0.114
CEBRACS8	<u>*0.122*</u>	**0.743**	0.147	0.240
CEBRACS9	**0.803**	<u>*−0.079*</u>	0.242	0.173
CEBRACS10	0.296	<u>*0.096*</u>	**0.661**	0.215
CEBRACS11	0.120	0.145	**0.644**	0.387
CEBRACS12	**0.840**	0.096	0.130	0.106
CEBRACS13	<u>*0.053*</u>	**0.877**	<u>*0.029*</u>	0.173
CEBRACS14	**0.794**	<u>*0.020*</u>	0.207	0.163
CEBRACS15	<u>*0.029*</u>	**0.907**	0.097	0.070
CEBRACS16	0.111	0.097	**0.814**	0.159
CEBRACS17	<u>*−0.047*</u>	**0.937**	0.112	0.050
CEBRACS18	−0.090	0.115	**0.833**	0.276
CEBRACS19	0.223	**0.435**	0.191	0.549
CEBRACS20	0.199	<u>*0.062*</u>	**0.786**	0.146
CEBRACS21	0.264	0.193	**0.608**	0.242
ω	0.973	0.948	0.939	
Alcohol Effects	-			
Bulimia	0.332	-		
Exercise	0.453	0.437	-	

CEBRACS = Compensatory Eating and Behaviors in Response to Alcohol Consumption Scale; λ = factor loadings; δ = Uniqueness; ω = McDonald's omega. All correlations significant at $p \leq .01$; Non significant cross-loadings are underlined and italicized

Gender invariance The fit indices of the 3-factor solution estimated separately in men and women (Models 2 − 1 and 2–2) are presented in Table 1. Results revealed a good fit to the data for men and women separately. Finally, results revealed that the 3-factor solution was completely invariant between men and women (Models 2–3 to 2–8).

DIF and latent factor means differences as function of age and BMI No significant improvement in model fit was found between saturated and factors-only models and the null model. Therefore, these results indicate a lack of DIF. Finally, no substantial improvement of model fit was observed between the saturated and the factors-only models, which suggests a lack of association between age and BMI, respectively, and the CEBRACS latent factors.

Discussion In Study 1, we adopted a data-driven approach to estimate the optimal factor structure of the Romanian CEBRACS. Our results showed that 3- and 4-factor models had good fit to the data, but parallel analysis suggested that we should retain a 3-factor model. Inspection of the two models also indicated that a high degree of cross-loadings in the 4-factor model, which were attenuated in the 3-factor model. As such, we elected to retain the 3-factor model for further analyses. This model retained two factors from Rahal et al. (2012), namely Alcohol Effects and Bulimia, and amalgamated two additional factors into a Diet and Exercise factor. The latter is consistent with the results of the EFA reported by Meinerding et al. (2024). In addition, the results of Study 1 also showed that the optimal 3-factor model was fully invariant across gender identity and showed a lack of DIF as a function of participant age and BMI.

Study 2

Participants

The initial sample consisted of 670 Romanian undergraduate students, but we excluded individuals who failed attention check items ($n=56$). The final sample of participants, therefore, included 614 individuals, of whom 375 identified as women and 239 identified as men. Participants ranged in age from 18 to 38 years ($M=21.51$, $SD=2.33$) and in BMI from 15.42 to 40.82 kg/m^2 ($M=23.02$, $SD=3.84$).

Materials

Food and alcohol disturbance Participants completed the 21-item Romanian translation of the CEBRACS (Rahal et al., 2012), as described above.

Symptoms of disordered eating To measure symptoms of disordered eating, we used the Body Image Screening Questionnaire for Eating Disorder Early Detection (BISQ; Jenaro et al., 2011; Romanian translation: Tomsa et al., 2012). This is a 24-item scale that measures symptoms of disordered eating along five dimensions (i.e., bulimia, anorexia, orthorexia, perception of obesity, and muscle dysmorphia). Items were rated on a 6-point scale (1 = *never*, 6 = *always*) and total scores have adequate reliability and discriminant validity (Tomsa et al., 2012). Composite reliability coefficients for subscale scores on the Romanian BISQ have been shown to be less-than-adequate. Therefore, a total score (i.e., the mean of all items) was estimated and used in the present study. Higher scores on this scale reflect greater symptoms of disordered eating. In the present study, McDonald's ω for the total score was 0.85 (95% CI = 0.84, 0.87).

Alcohol consumption and behaviours To assess alcohol consumption and related behaviours, we used the Alcohol Use Disorders Identification Test (AUDIT; Babor et al., 2001; Saunders et al., 1993; Romanian translation: Jianu, 1996). The AUDIT is a 10-item instrument that assesses three domains of alcohol use, namely alcohol consumption (frequency, typical quantity, and heavy episodic drinking), alcohol dependence (impaired control, failure to meet expectation, and morning drinking), and negative consequences (guilt, blackouts, injuries, and others' concerns). Each item is scored between 0 and 4 points, with five response options for the first eight items and three response options for the final two items. Although there is some indication that the AUDIT is best conceptualised using a 2-factor structure (assessing alcohol use and problems, respectively; Horváth et al., 2023), scores on the Romanian version of the AUDIT have been found to have a unidimensional factor structure (Jianu, 1996). As such, we computed an overall AUDIT score, with higher scores reflecting more problematic or hazardous alcohol use. In the current study, McDonald's ω for the total score was 0.87 (95% CI = 0.85, 0.87).

Body appreciation We measured body appreciation, a facet of positive body image, using the Body Appreciation Scale-2 (BAS-2; Tylka & Wood-Barcalow, 2015; Romanian translation: Swami et al., 2017). This is a 10-item scale that assesses acceptance of one's body, respect and care for one's body, and protection of one's body from unrealistic beauty standards. Items were rated on a 5-point scale, ranging from 1 (*never*) to 5 (*always*) and an overall score was computed as the mean of all items, with higher scores indicating greater body appreciation. Scores on the Romanian version of the BAS-2 have been shown to have a unidimensional factor structure (Swami et al., 2023b) with adequate composite reliability, construct validity, and test-retest reliability (Swami et al., 2017). In the present study, McDonald's ω for BAS-2 scores was 0.96 (95% CI = 0.96, 0.97).

Self-esteem Self-esteem was assessed using the Rosenberg Self-Esteem Scale (RSES; Rosenberg, 1965; Romanian translation: Schmitt & Allik, 2005), a 10-item measure of global self-evaluations of worth as a human being. All items were rated on a 4-point scale (1 = *definitely disagree*, 4 = *definitely agree*) and an overall score was computed as the mean of all 10 items, following reverse-coding of five items. The Romanian version of the RSES has evidenced factorial validity (Schmitt & Allik, 2005). In the present study, McDonald's ω for this scale was 0.91 (95% CI = 0.89, 0.92).

Life satisfaction Participants were asked to complete the 5-item Satisfaction with Life Scale (SLS; Diener et al., 1985; Romanian translation: Stevens et al., 2012), a widely used self-report measure of subjective well-being. Items on this scale were rated on a 7-point scale (1 = *strongly disagree*, 7 = *strongly agree*). An overall score was computed at the mean of all five items, so that higher scores reflect greater life satisfaction. Scores on the Romanian version of the scale have been shown to have a unidimensional factor structure and evidence strong patterns of convergent validity (Stevens et al., 2012; Swami et al., 2025). In the present study, McDonald's ω for this scale was 0.86 (95% CI = 0.84, 0.88).

Demographics Participants were asked to self-report their height, weight, age, and gender identity. Participant BMI was calculated as kg/m^2.

Procedures

All procedures were identical to that of Study 1, with the exception that recruitment took place in May 2025. The instruments within the survey were presented in a fixed order.

(CEBRACS, BISQ, BAS-2, RSES, SLS, and AUDIT).

Statistical analyses

Analyses were performed with the same software and estimator used in Study 1. No missing data were observed for

CEBRACS responses. Firstly, the a priori 3-factor model retained in Study 1 was estimated using CFA and ESEM. In the CFA model, the CEBRACS ratings were explained by three correlated latent factors and no-cross-loadings were estimated. In the ESEM model, all cross-loadings were estimated using a confirmatory oblique target rotation procedure (Asparouhov & Muthén, 2009).

Fit indices used were identical to those reported in Study 1. Gender invariance, DIF, and latent mean differences as a function of age and BMI were estimated for the CFA and ESEM models following the same sequences reported in Study 1. Finally, analysis of construct validity was conducted using a structural equation model (SEM) in which the CEBRACS latent factor structure was estimated based on the model providing optimal representation of the data (CFA or ESEM). In the SEM, all measures were correlated (i.e., the CEBRACS latent factors and the observed scores of all additional measures).

Table 3 Standardised parameters estimates from the 3-Exploratory structural equation modelling representation of the CEBRACS in the sample from study 2

Items	Alcohol Effects (λ)	Bulimia (λ)	Diet & exercise (λ)	δ
CEBRACS1	**0.885**	−0.217	0.198	0.164
CEBRACS3	**0.973**	−0.310	0.222	0.044
CEBRACS6	**0.868**	−0.144	0.231	0.086
CEBRACS7	**0.890**	0.149	*−0.054*	0.111
CEBRACS9	**0.805**	0.334	−0.131	0.124
CEBRACS12	**0.830**	0.215	*−0.024*	0.099
CEBRACS14	**0.847**	0.353	−0.179	0.082
CEBRACS5	0.213	**0.721**	*0.042*	0.211
CEBRACS8	0.123	**0.807**	*0.079*	0.121
CEBRACS13	0.174	**0.837**	*−0.012*	0.124
CEBRACS15	*−0.101*	**0.748**	0.343	0.106
CEBRACS17	*0.000*	**0.833**	0.188	0.066
CEBRACS19	*0.113*	**0.468**	0.405	0.235
CEBRACS2	*0.054*	*−0.010*	**0.776**	0.349
CEBRACS4	0.496	*−0.058*	**0.532**	0.185
CEBRACS10	0.199	*0.026*	**0.728**	0.204
CEBRACS11	*0.064*	0.200	**0.647**	0.298
CEBRACS16	−0.078	0.113	**0.917**	0.110
CEBRACS18	−0.137	*0.086*	**0.864**	0.303
CEBRACS20	*−0.019*	0.156	**0.847**	0.109
CEBRACS21	*0.057*	0.289	**0.649**	0.177
ω	0.981	0.958	0.953	
Alcohol Effects	-			
Bulimia	0.556	-		
Diet & Exercise	0.673	0.657	-	

CEBRACS = Compensatory Eating and Behaviors in Response to Alcohol Consumption Scale; λ = factor loadings; δ = Uniqueness; ω = McDonald's omega. All correlations significant at $p \leq .01$; Non significant cross-loadings are underlined and italicized

Results and discussion

CFA and ESEM representations Results from the CFA and ESEM solutions are presented in Table 1. The 3-factor CFA solution (Model 4−1) showed acceptable-to-good fit (CFI and TLI > 0.95 and RMSEA < 0.08). In contrast, the 3-factor ESEM solution (Model 4−2) had a good fit to the data (CFI and TLI > 0.95 and RMSEA < 0.06). Table S2 in the Supplementary Materials and Table 3 presents the standardized parameters and composite reliability coefficients for the CFA and ESEM solutions, respectively.

In the CFA solution (see Table S2 in the Supplementary Materials), the factor loadings of the CEBRACS items were satisfactory (|λ| = 0.782–0.956.782.956, M|λ| = 0.907). The latent factors demonstrated excellent composite reliability coefficients (ω = 0.962 to 0.980, M_{ω} = 0.971) and their correlations (Mr = 0.782) were significant, positive, and high. Table 3 shows that the factors from the ESEM solution were also satisfactory (|λ| = 0.468–0.973.468.973, M|λ| = 0.784). The latent factors demonstrated excellent composite reliability coefficients (ω = 0.953 to 0.981, M_{ω} = 0.964), and their correlations were reduced (r = .629). This suggests that the lower level of overlap between factors could in fact be explained by the presence of cross-loadings (M|λ| = 0.157). Based on these results, the 3-factor ESEM model was retained for the subsequent analyses.

Gender invariance The fit indices of the 3-factor ESEM solution estimated separately in men and women (Models 5−1 and 5−2) are presented in Table 1. Results revealed good fit to the data for men and women. Finally, results revealed that the 3-factor ESEM solution was completely invariant between men and women (Models 5−3 to 5–8).

DIF and latent factor means differences as function of age and BMI Model fit did not improve significantly between saturated and factors-only models and the null model. Therefore, these results provide support for a lack of DIF. Finally, no substantial improvement of model fit was observed between the saturated and the factors only models. This result suggests a lack of association between age and BMI, respectively, and the CEBRACS latent factors.

Construct validity Fit indices from the SEM were acceptable: $\chi^2(240) = 484.467$, CFI = 0.993, TLI = 0.991, RMSEA = 0.041 (90% CI = 0.035, 0.046). Results presented in Table 4 showed that the three factors of the CEBRACS were significantly and (a) positively correlated with symptoms of disordered eating and alcohol consumption and

Table 4 Construct validity analyses from the Three-Factor exploratory structural equation modelling representation of the CEBRACS in sample from study 2

	Alcohol Effects	Bulimia	Diet & Exercise
Self-esteem	−0.211*	−0.211*	−0.191*
Symptoms of disordered eating	0.267*	0.356*	0.445*
Body appreciation	−0.212*	−0.157*	−0.243*
Life satisfaction	−0.251*	−0.202*	−0.187*
Problematic alcohol use	0.501*	0.269*	0.327*

CEBRACS = Compensatory Eating and Behaviors in Response to Alcohol Consumption Scale; * $p \leq .01$

behaviours; and (b) negatively correlated with self-esteem, body appreciation, and life satisfaction.

Discussion The results of Study 2 showed that both the CFA- and ESEM-based, 3-factor models of the CEBRACS had adequate fit to our data. Of the two models, however, the ESEM model had better fit indices and lower factor inter-correlations. In fact, mean factor inter-correlations in the CFA model was very high, suggesting a high degree of nomological overlap and problematic item cross-loadings. Based on these results, we selected the ESEM model for further analyses and this model was found to be completely invariant across gender identity and showed a lack of DIF as a function of participant age and BMI. In Study 2, we were also able to demonstrate adequate evidence of convergent and concurrent validity, insofar as the CEBRACS factors were positively associated with symptoms of disordered eating and problematic alcohol use, and negatively correlated with body appreciation and life satisfaction.

General discussion

A Romanian translation of the CEBRACS was evaluated for its psychometric properties in the present study. Overall, the results of our EFA-to-CFA/ESEM analytic strategy showed that it was possible to retain a 3-factor model of the CEBRACS with all 21 items. This was the model that showed optimal factorial validity in a data-driven approach (i.e., the EFA in Study 1) and in our cross-validation (i.e., the CFA and ESEM in Study 2). Additionally, this 3-factor model achieved complete measurement invariance across gender identity and showed a lack of DIF as a function of participant age and BMI in both studies. Finally, the three factors of the Romanian CEBRACS also evidenced adequate convergent and concurrent validity, insofar as associations with additional constructs were all in hypothesised directions. Here, we evaluate these results in greater detail before considering the future of the CEBRACS.

In terms of factorial validity, our results suggest that it is possible to extract a 3-factor model of the CEBRACS in our data. This model retains two factors from the original conceptualisation of the CEBRACS (Rahal et al., 2012), namely the Alcohol Effects and Bulimia factors, whereas a third factor was an amalgamation of two factors from the parent study (Diet and Exercise). Broadly speaking, our 3-factor model is similar to the 3-factor model extracted via EFA in a sample of participants from the United States (Meinerding et al., 2024). Notably, however, Meinerding et al. (2024) also found that their 3-factor model had poor CFA-based fit. In contrast, our cross-validation results in Study 2 showed that the 3-factor model had adequate fit when recovered using both CFA and ESEM, although it was the latter that had superior fit.

It is possible that Meinerding et al.'s (2024) failure to find adequate fit for their 3-factor model reflects the fact that CFA only allows items to load onto the theoretically specified latent factors (Morin, 2023; Swami et al., 2023a). However, the items of the CEBRACS appear to cross-load to a substantive degree – in the parent study by Rahal et al. (2012), for instance, at least one item cross-loaded at 0.20 and 11 items cross-loaded at about 0.10. Indeed, even negligible cross-loadings (i.e., as low as 0.10) may sometimes exert salient effects on factor modelling (Asparouhov et al., 2015) and failure to account for this may have led to biases model specification (Marsh et al., 2020). In contrast, ESEM allows for cross-loadings and thus typically provides improved and more accurate representations of data than CFA-based models (Swami et al., 2023a).

However, one aspect of the existing literature that is more difficult to reconcile relates to the variability of factor structures that have been extracted. For example, based on EFAs, previous studies have recommended extraction of 2-factor models (Choquette et al., 2020), a 4-factor model that deviates substantively from the parent model (Ritz et al., 2023), and a 5-factor model (Pinna et al., 2015). It is possible that these discrepancies reflect sampling differences (e.g., variations in age or other demographic characteristics), the impact of culture (e.g., attitudes towards alcohol consumption) on conceptualisations of FAD, or variations in the specific type of factor analysis that is used (e.g., principal components analysis in the parent study versus EFA in subsequent studies). While further research is necessary to understand the root of such discrepancies, a practical outcome is that it may be very difficult to consider cross-national or cross-linguistic differences in CEBRACS scores given the existing factor structure non-equivalence.

Beyond factorial validity, the results of our study also showed that the 3-factor model of the CEBRACS evidenced full measurement invariance across gender identity. This finding is consistent with the parent study (Rahal et al.,

2012) and other studies (e.g., Peralta & Barr, 2017), where no significant gender differences in (observed or manifest) CEBRACS factor scores have been reported. It is also consistent with the broader literature showing non-significant gender differences in FAD, irrespective of the specific instrument utilised (Speed et al., 2022). Additionally, our results also showed that there was a lack of DIF as a function of participant age and BMI. That is, participant age and BMI did not appear to have any substantive impact on the way in which the items of the Romanian CEBRACS were understood and completed. Taken together, these results suggest that scholars wishing to use the Romanian CEBRACS can be confident that participant characteristics – gender identity, age, and BMI – do not unduly bias responding in ways that may impact model specification.

Finally, the results of Study 2 showed that the Romanian CEBRACS evidenced good patterns of construct validity. More specifically, and consistent with current definitions of FAD (Berry et al., 2024; Choquette et al., 2018b), we found strong evidence of convergent validity insofar as higher CEBRACS latent means were associated with greater symptoms of disordered eating and more problematic alcohol use. In broad outline, these findings are important in their own right because they support suggestions of an overlap between alcohol abuse and eating disorders (Berry et al., 2024). Additionally, we also found evidence of adequate concurrent validity, insofar as higher CEBRACS latent means were significantly associated with lower body appreciation and psychological well-being (i.e., self-esteem and life satisfaction). These results are consistent with previous work showing that CEBRACS scores are associated with body image (e.g., Griffin & Vogt, 2021; see also Thompson, 2025) and poorer psychological well-being (e.g., Laghi et al., 2020).

Nevertheless, there are several limitations of the present work that should be considered alongside our results. First, in considering the factor structure of the CEBRACS, we did not take into account the time-specific aspect of the instrument. While this is consistent with the intention of the developers (Rahal et al., 2012), it is possible that alternative models would have emerged had we conducted analyses separately as a function of time (i.e., before, during, and after drinking; Choquette et al., 2020). In a similar vein, we did not consider fitting a bifactor model given that an overall composite score or G-factor is unlikely to have strong face validity and would likely lack interpretability (i.e., because items aggregate behaviours across time and motive). There was some evidence in the present work to support this thinking, wherein the unidimensional model of the CEBRACS had poor fit in Study 1.

In terms of our study design, we caution that our sample should not be considered representative of the wider Romanian population. Indeed, because we recruited university students, it is also unlikely that our samples would be representative of young adults in Romania, where under a quarter of young adults have a tertiary degree (European Commission, 2025). Moreover, there are large inter- and intra-regional differences in socioeconomic status, rural-urban residence, and understandings of health and illness across Romania (e.g., Vintilă et al., 2020; Voinea et al., 2019), all of which may have impacted our findings. As such, it will be important in future work to determine to what extent our findings are replicable in more representative samples of Romanian university students. It may also be useful to consider the extent to which the Romanian CEBRACS retains its psychometric properties in the wider Romanian population (e.g., Moeck & Thomas, 2021) or in specific subpopulations, such as student athletes (e.g., Galante et al., 2017). In future work, it may also be useful to assess associations between CEBRACS latent factor scores and additional constructs, such as indices of negative body image, eating patterns or behaviours, and other compensatory behaviours (Sedemedes et al., 2023). Notably, the present study did not include assessment of divergent validity and temporal stability of scores, which could be rectified in future work. [1]

To conclude, the present study suggests that the Romanian version of the CEBRACS can be conceptualised as a 3-factor instrument. This model of the CEBRACS evidenced good composite reliabilities, measurement invariance across gender identity, and construct validity, and lacked DIF across participant age and BMI. As such, scholars and practitioners wishing to use the CEBRACS with Romanian-speaking populations can do so with the knowledge that this translation demonstrates adequate psychometric properties. However, this does not detract from calls for better instruments to measure the construct of FAD (Berry et al., 2024), in keeping with recent reconceptualisations of the construct (e.g., Thompson, 2025). In fact, if we consider it important to have an instrument that can be deployed across national, culture, or linguistic

[1] Because total CEBRACS scores have uncertain face validity (i.e., it is unclear what a total score assesses in terms of latent meaning), we elected not to assess fit of a bifactor model.

boundaries to assess FAD, then we will certainly need a measure other than the CEBRACS.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethical approval All procedures performed in this study involving human participants were in accordance with the ethical standards of the institutional research committee of Alexandru Ioan Cuza University and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. This article does not contain any studies with animals performed by any of the authors.

Consent to publish Participants consent to have their anonymized data published in a journal article.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Conflict of interest The other authors declare that they have no conflict of interest.

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